

Tech Talks South Tyrol

Infrastructure as code with automated credentials management using 1Password

Outline

1. Our old setup
2. Our new setup
3. 1Password Connect
4. Terraform - Secrets Management
5. Kubernetes - Secrets Management

Our old setup

Our old setup

Infrastructure Setup:

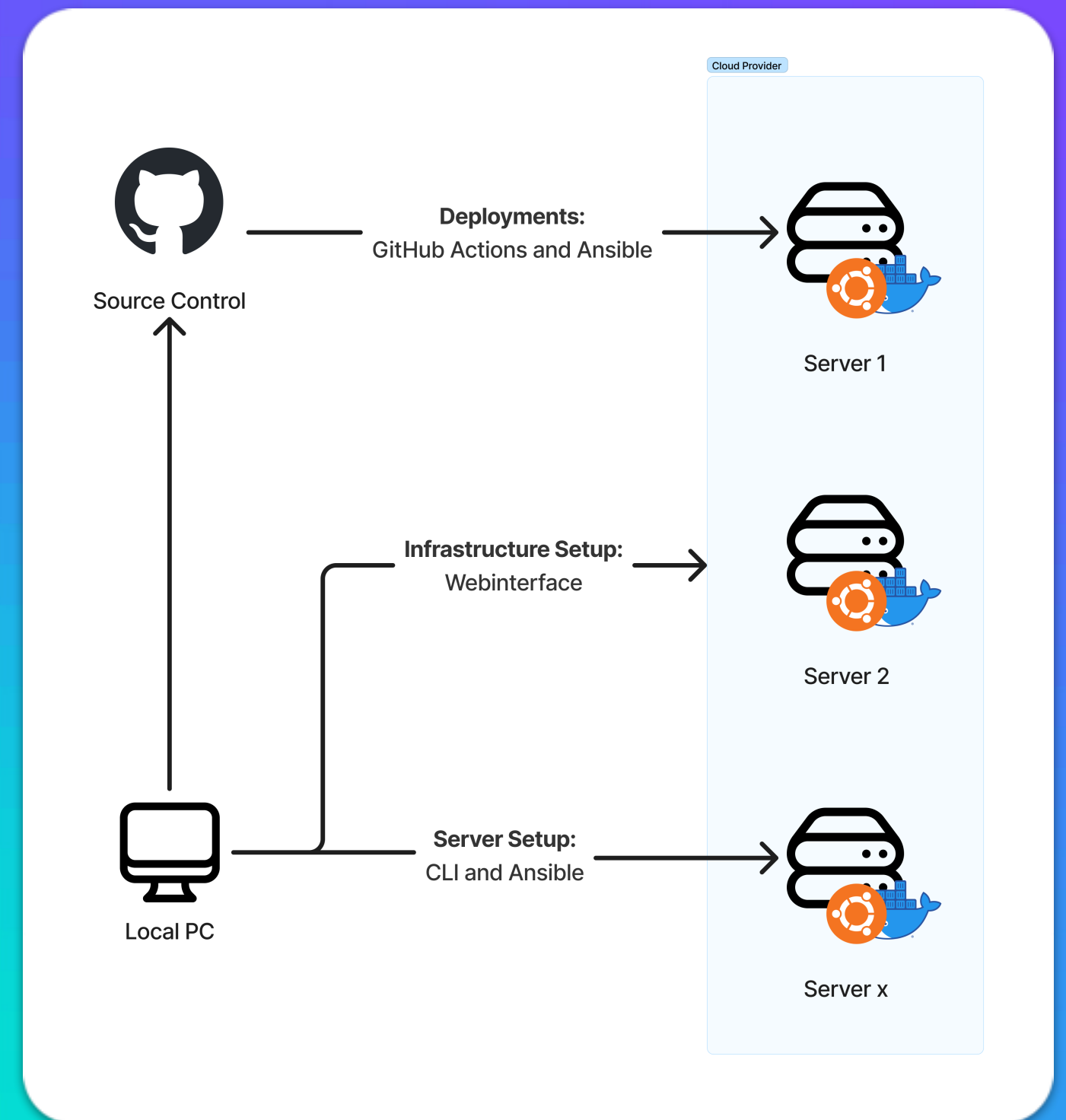
- Manually using web interface

Server Setup:

- Stored in a central Git repository
- Executed locally
- Using Ansible Playbooks

Deployments:

- Stored in every project Git repository
- Executed using GitHub Actions
- Using Ansible Playbooks



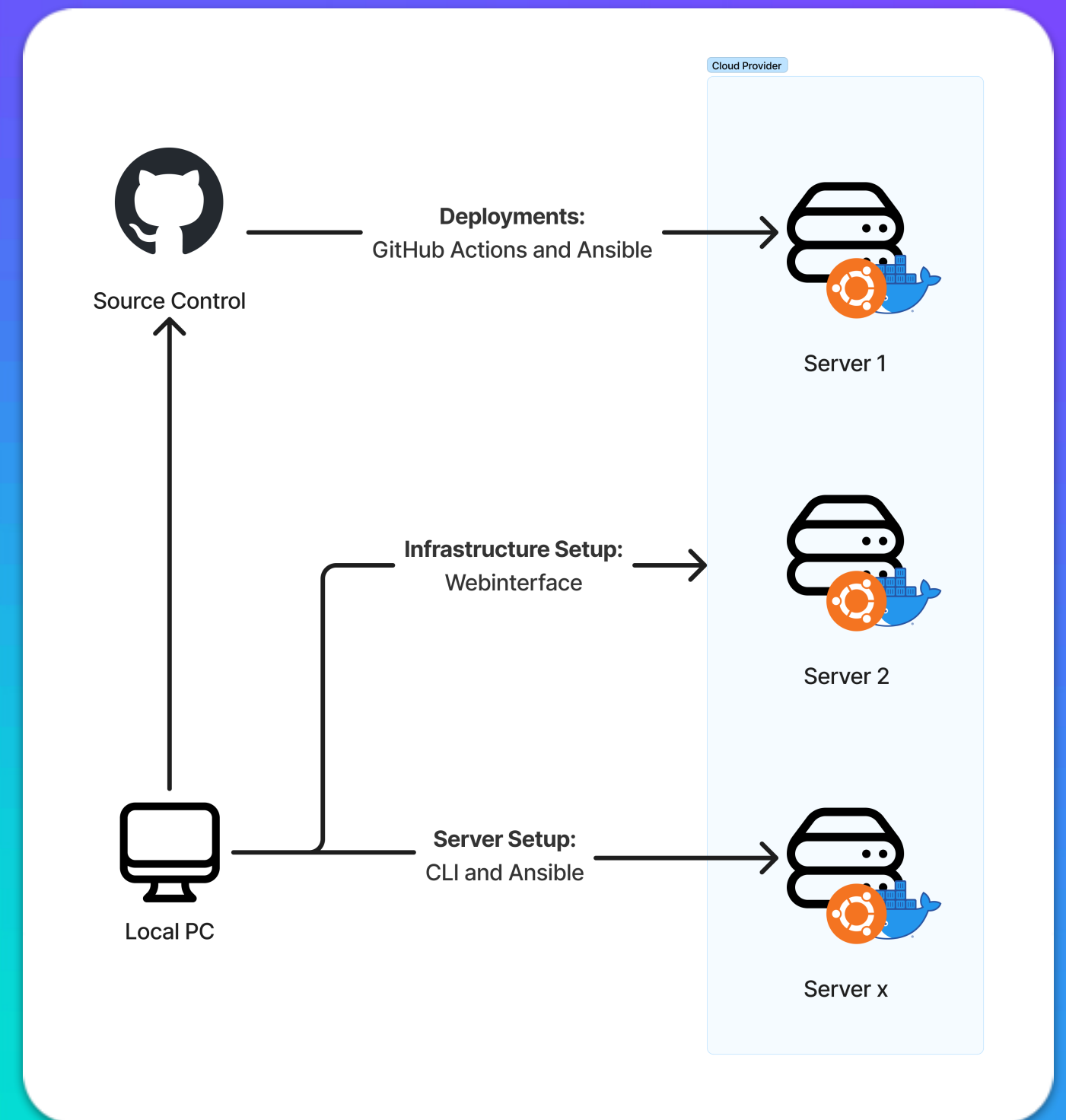
Our old setup

Pros:

- + Easy to setup and use
- + Testable on local environment
- + Incremental updates for the single servers

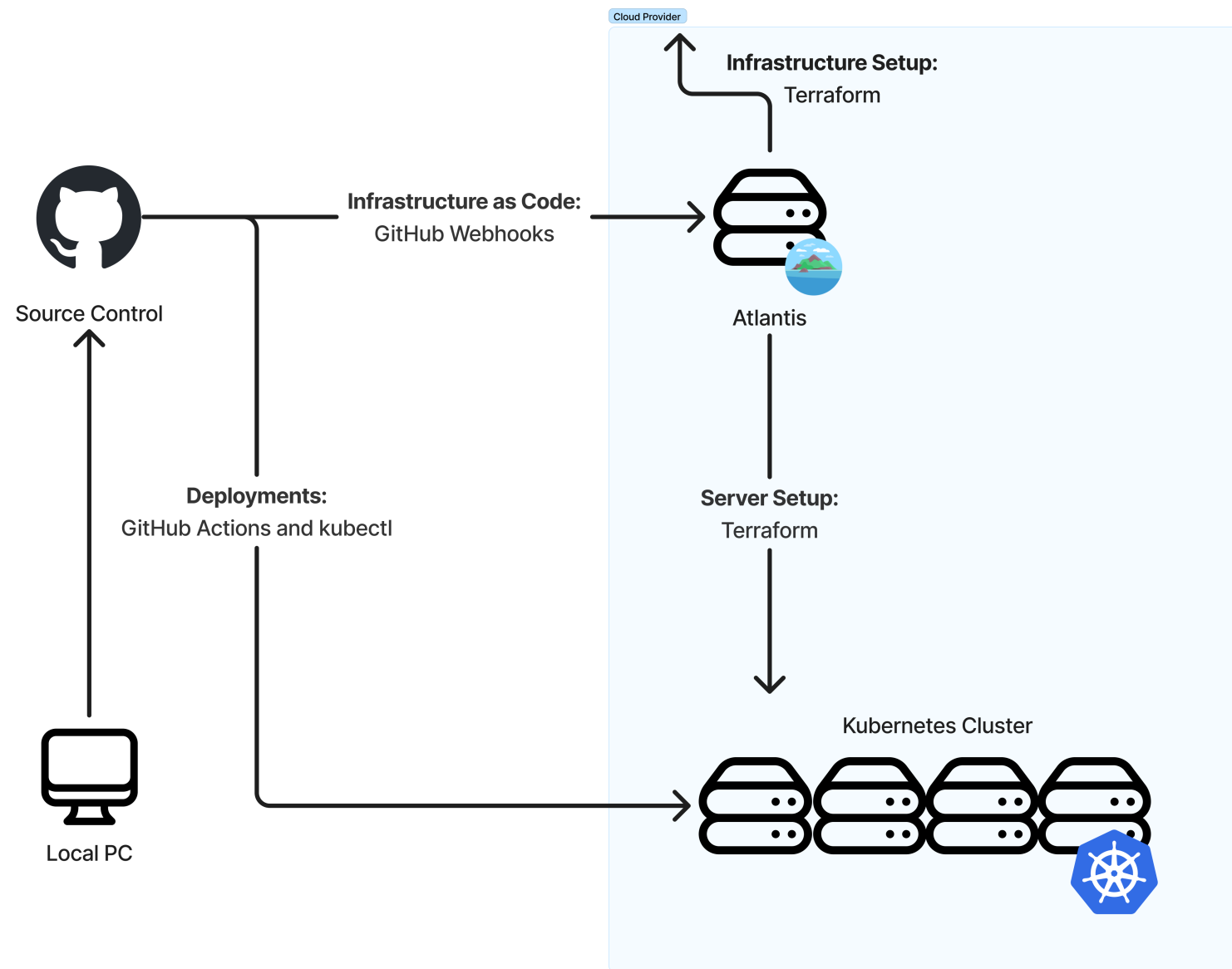
Cons:

- No synced state of code and infrastructure
- No green/blue deployments
- Hard to scale
- Secrets stored in GitHub Secrets



Our new setup

Our new setup



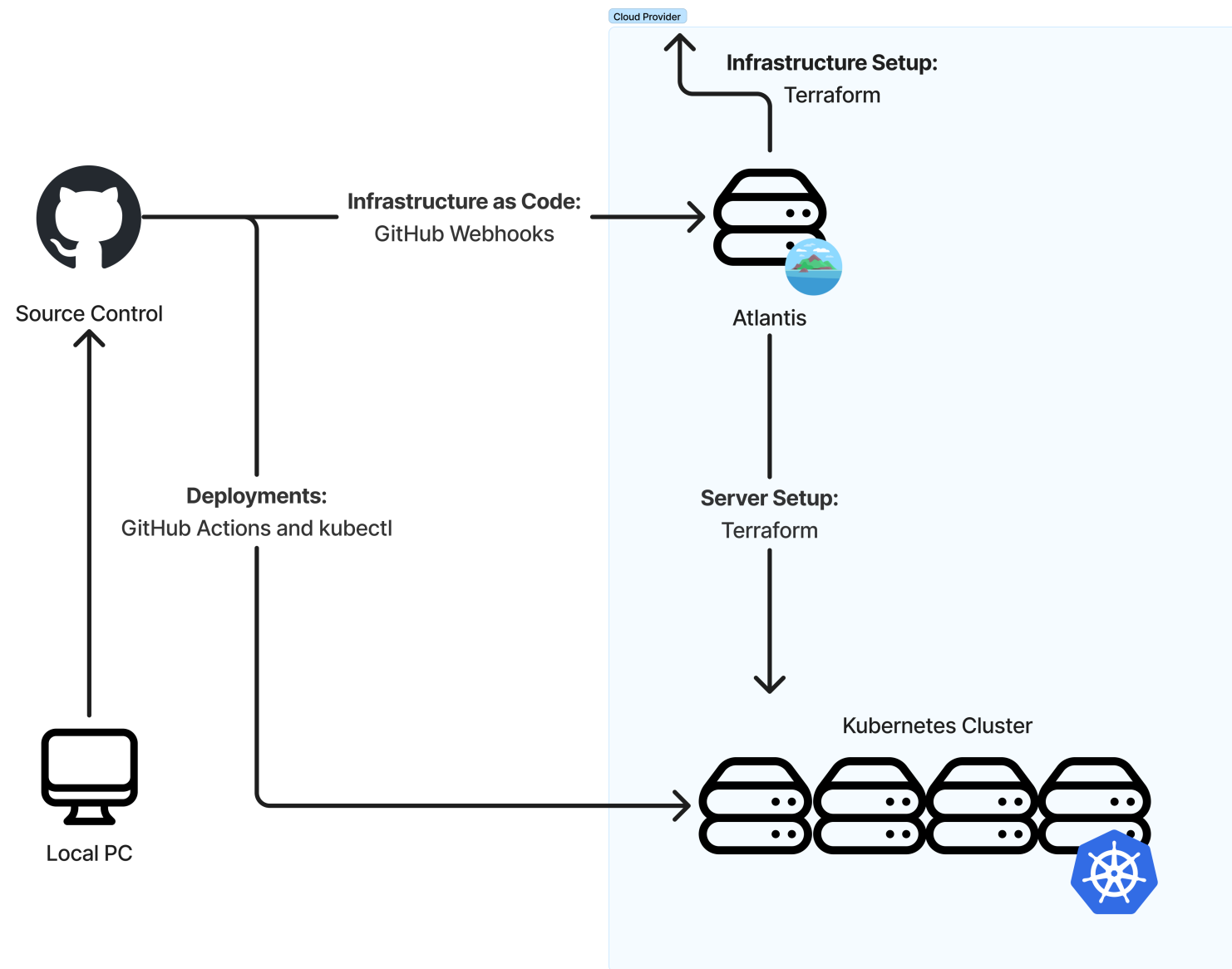
Infrastructure / Server Setup:

- Planning triggered using GitHub Webhooks
- Execution triggered using PR comments
- Executed by Atlantis using Terraform

Deployments:

- Executed by GitHub Actions using kubectl commands

Our new setup



Infrastructure as Code

- + Increased efficiency
- + Reproducibility
- + Consistency
- + Version Control
- + Collaboration
- + Scalability

Kubernetes

- + Green/blue deployments
- + Scalability
- + No required adaptations for our applications

What about secrets?

What about secrets?

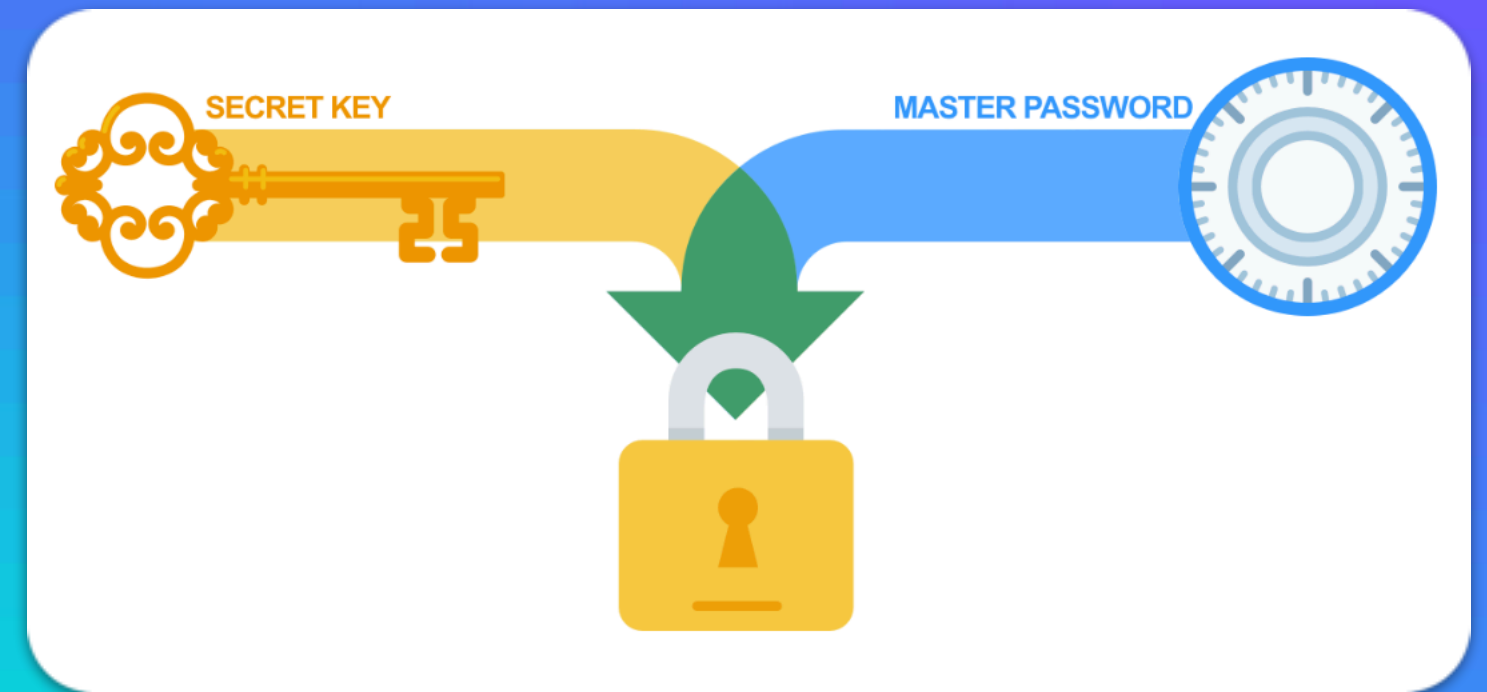
- How can we provide secrets to Atlantis and Terraform for setting up the infrastructure?
- How can we provide secrets to a deployment of an application inside Kubernetes?
- How can we ensure a secure strategy for our secrets?
 - Secure Storage
 - Access Control
 - Encryption
 - Automation



1Password Connect

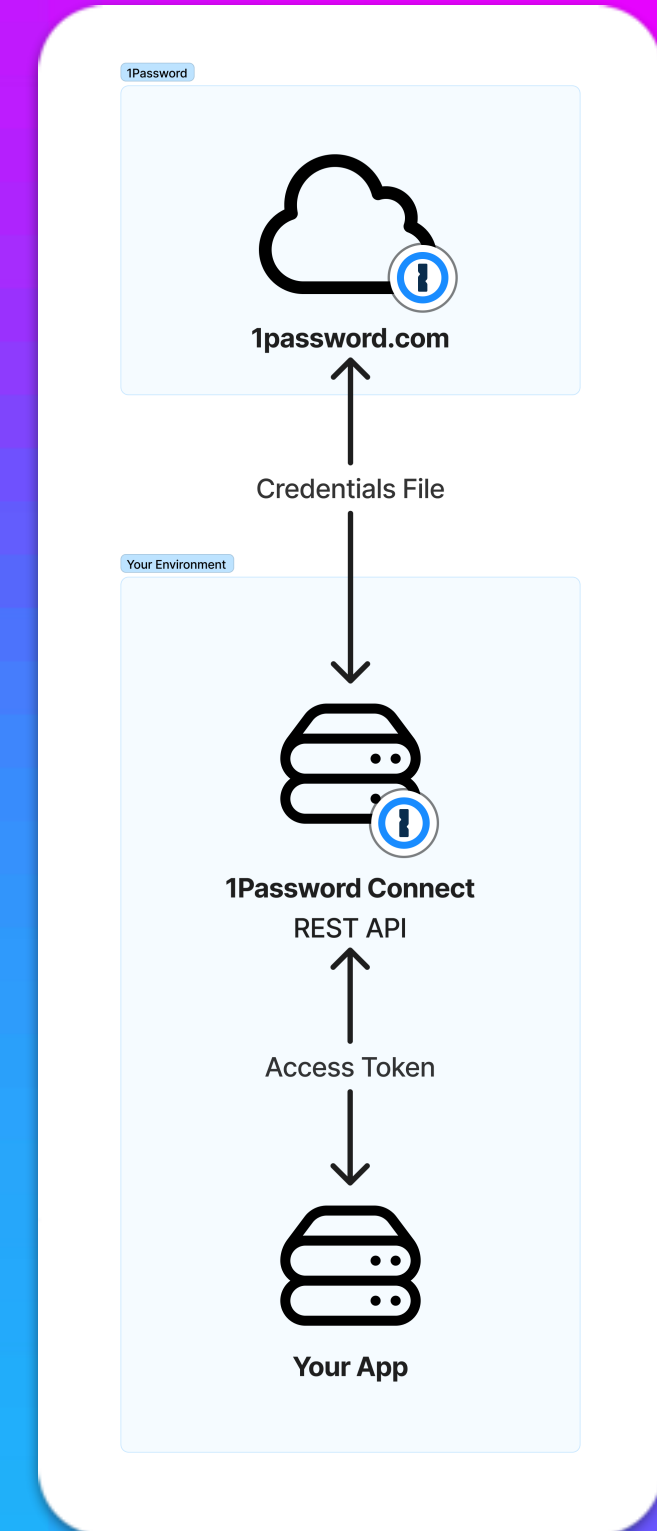
1Password

- A secure and encrypted password manager
- It allows users to store and manage passwords and other sensitive data in a vault
- The platform uses end-to-end encryption, ensuring only the user can access their data
- White Paper: <https://1passwordstatic.com/files/security/1password-white-paper.pdf>



1Password Connect

- Provides REST API to securely access data from your 1Password account
- Access of the Connect server to the 1Password Cloud is secured by a credentials file
- Access of the application to the Connect server is secured by access tokens



Why is there a dedicated server required?

- **Key handling:**

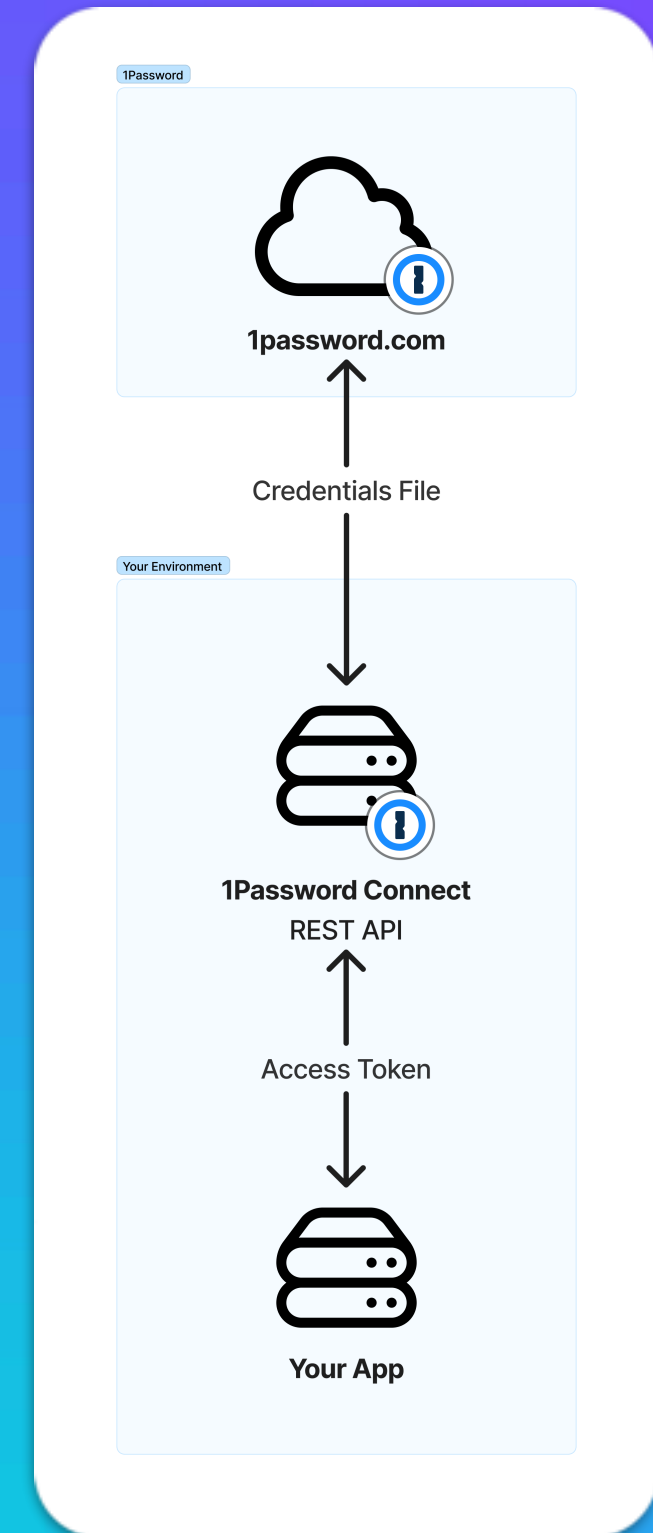
Under the hood the secret key and master password are only used to decrypt other keys.

- **Rate limiting:**

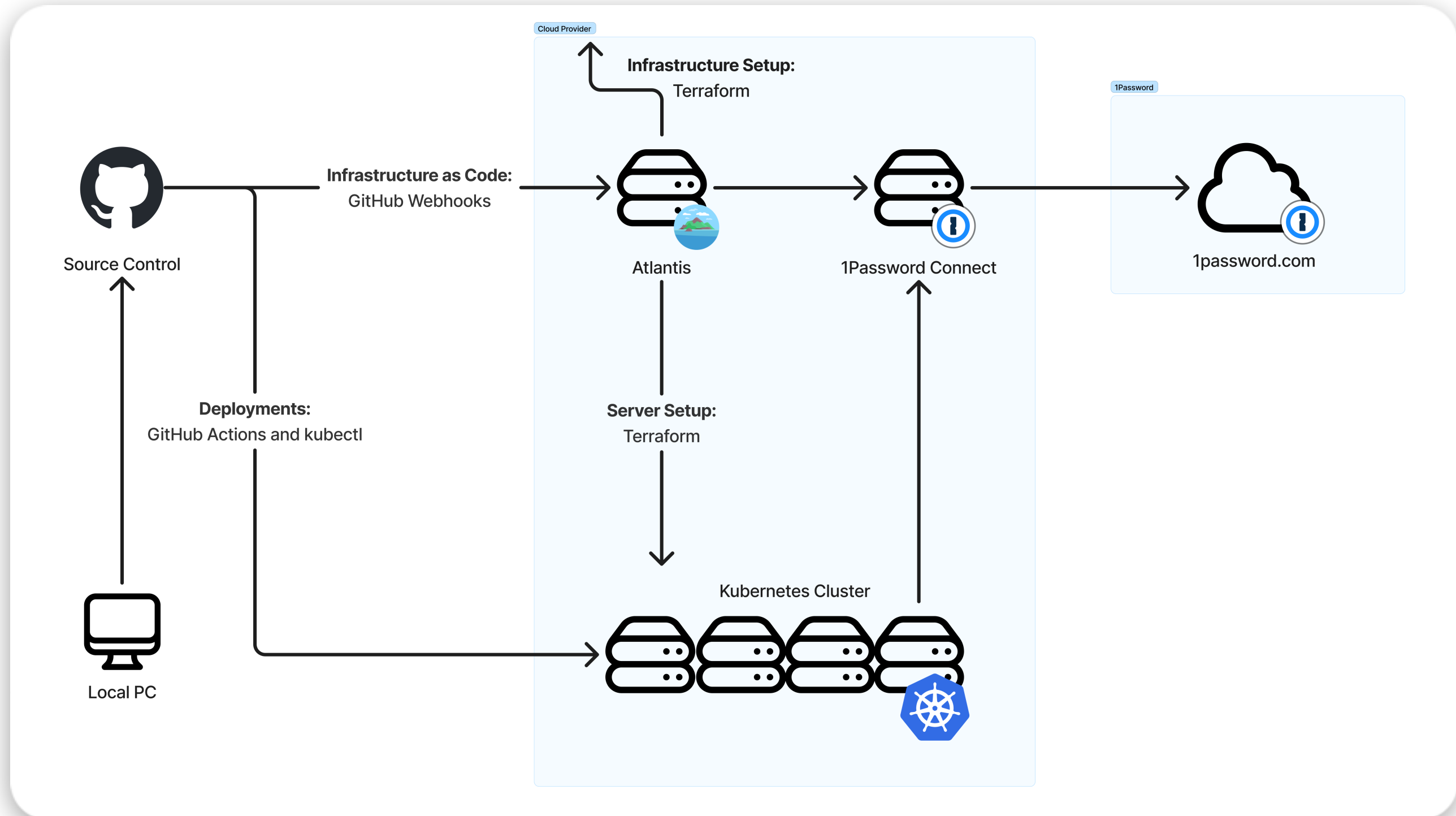
1Password cloud has very strict rate limiting. The Connect server operates as cache.

- **Reliability:**

The connect server can still respond to subsequent calls, even if the 1Password cloud is offline.



1Password Connect



1 Password Integrations

Access Tokens:
Permissions of access token
to different vaults

Integration

Integrations

ActiveDirectoryPermissions

0

Alex Lanz

Dashboard

People

Invitations

Groups

Vaults

Integrations

Billing

Settings

Give Feedback

View your vault items

You'll find vaults and saved items here

Infrastructure

Secrets Automation

Sync activity:

2023-05-16 at 8:11:50 PM +02:00

Vault access credits: 5

Edit Details

Version

1.6.1

More Actions

Access Tokens

New Token

Atlantis	
Kubernetes - Test	
Expiration Never	
Vaults Kubernetes - Test	Read
Kubernetes - Prod	

Vaults

Manage

Atlantis - Test	Read	
Atlantis - Prod	Read	
Kubernetes - Prod	Read	

Vaults:
Permissions of integration
to different vaults

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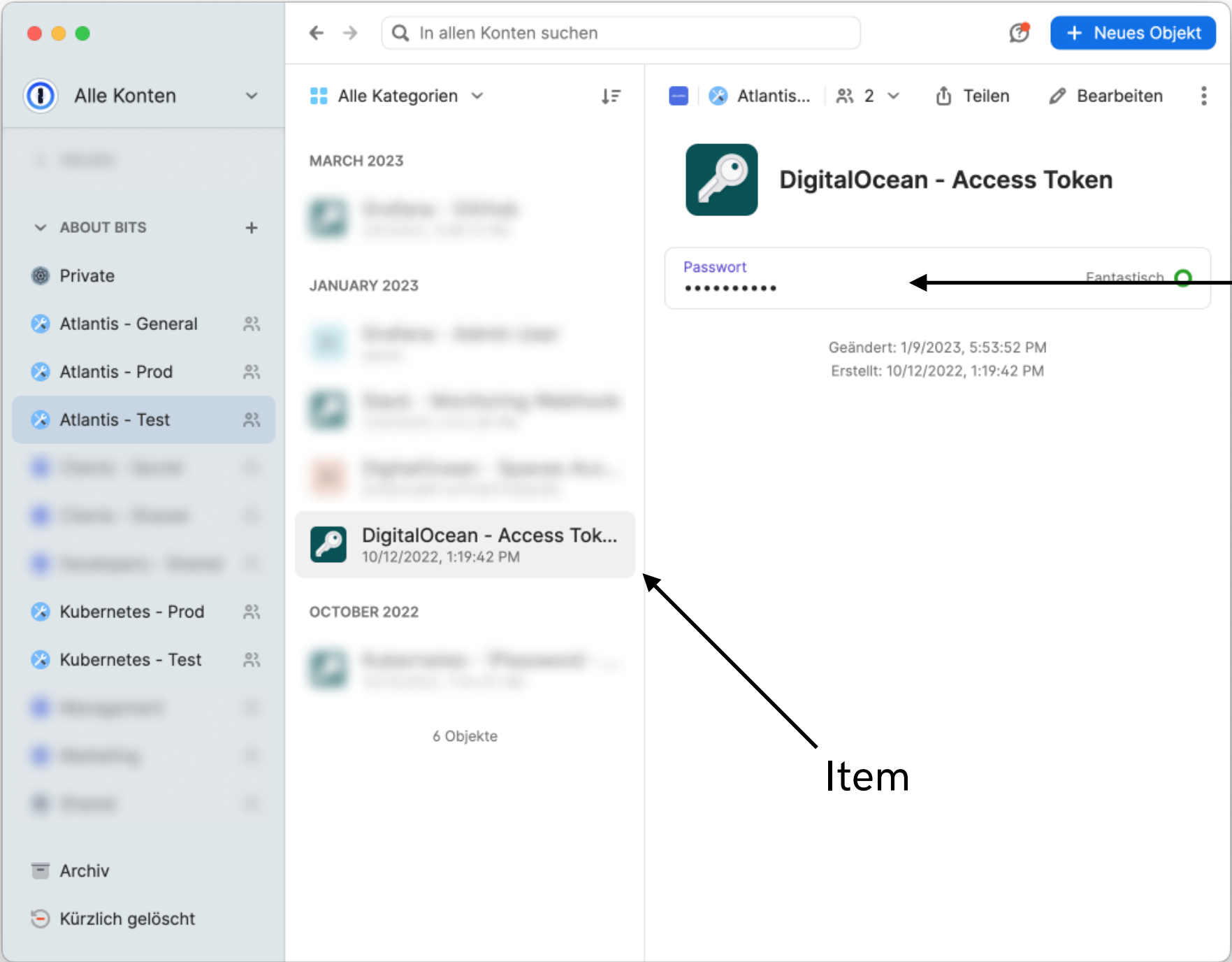
16

Terraform

Secrets Management

Terraform - Secrets Management

Vault



Field

Item

Terraform - Secrets Management

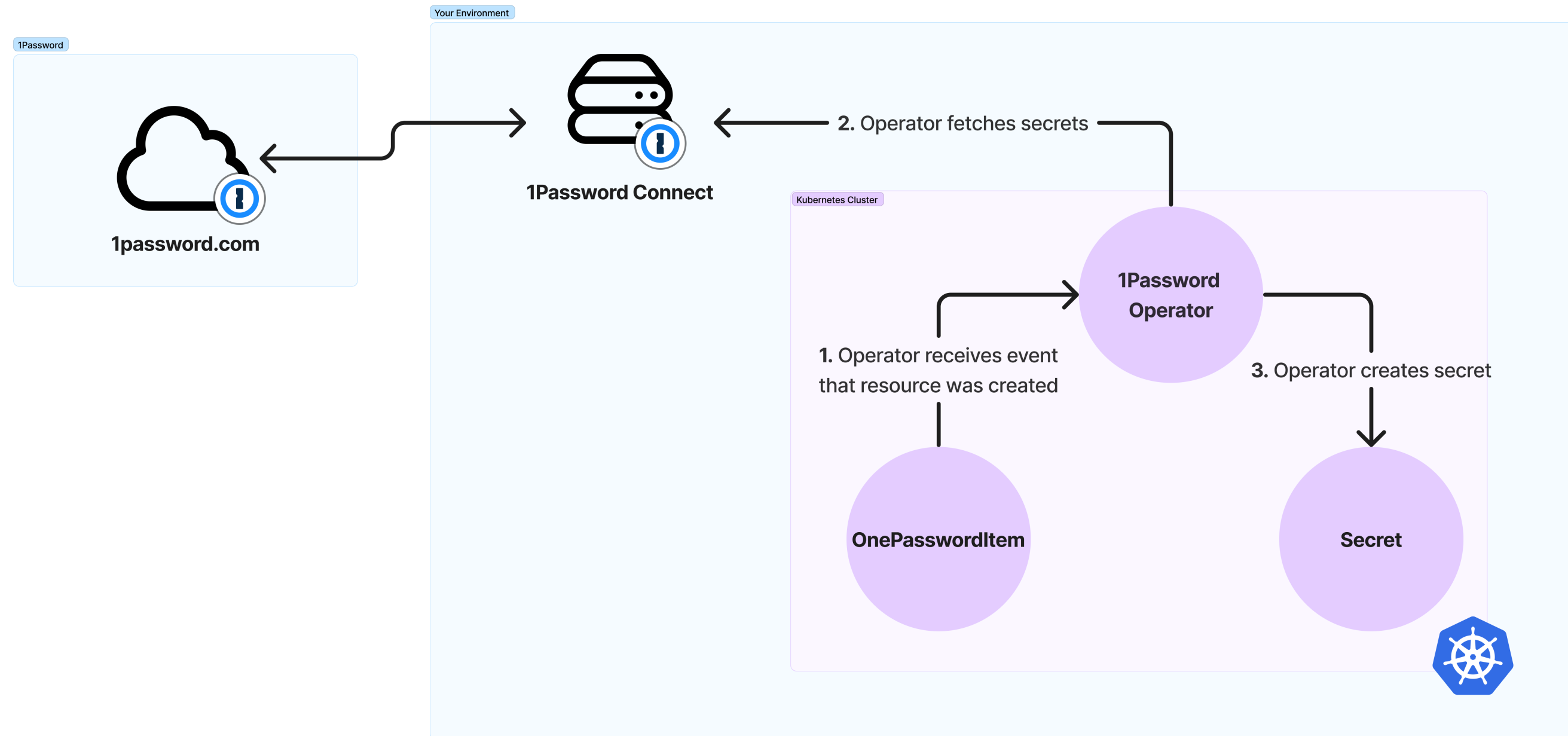
```
provider "onpassword" {  
  url = "https://1password-connect.example.com"  
  token = "xxx"  
}  
  
data "onpassword_vault" "atlantis_vault" {  
  name = "Atlantis - Test"  
}  
  
data "onpassword_item" "digitalocean_access_token" {  
  vault = data.onpassword_vault.atlantis_vault.uuid  
  title = "DigitalOcean - Access Token"  
}  
  
provider "digitalocean" {  
  token = data.onpassword_item.digitalocean_access_token.password  
}
```

- Official provider: <https://registry.terraform.io/providers/1Password/onpassword>
- Only one single secret required to be stored in Atlantis
- Secrets from 1Password can be used to set up other providers

Kubernetes Secrets Management

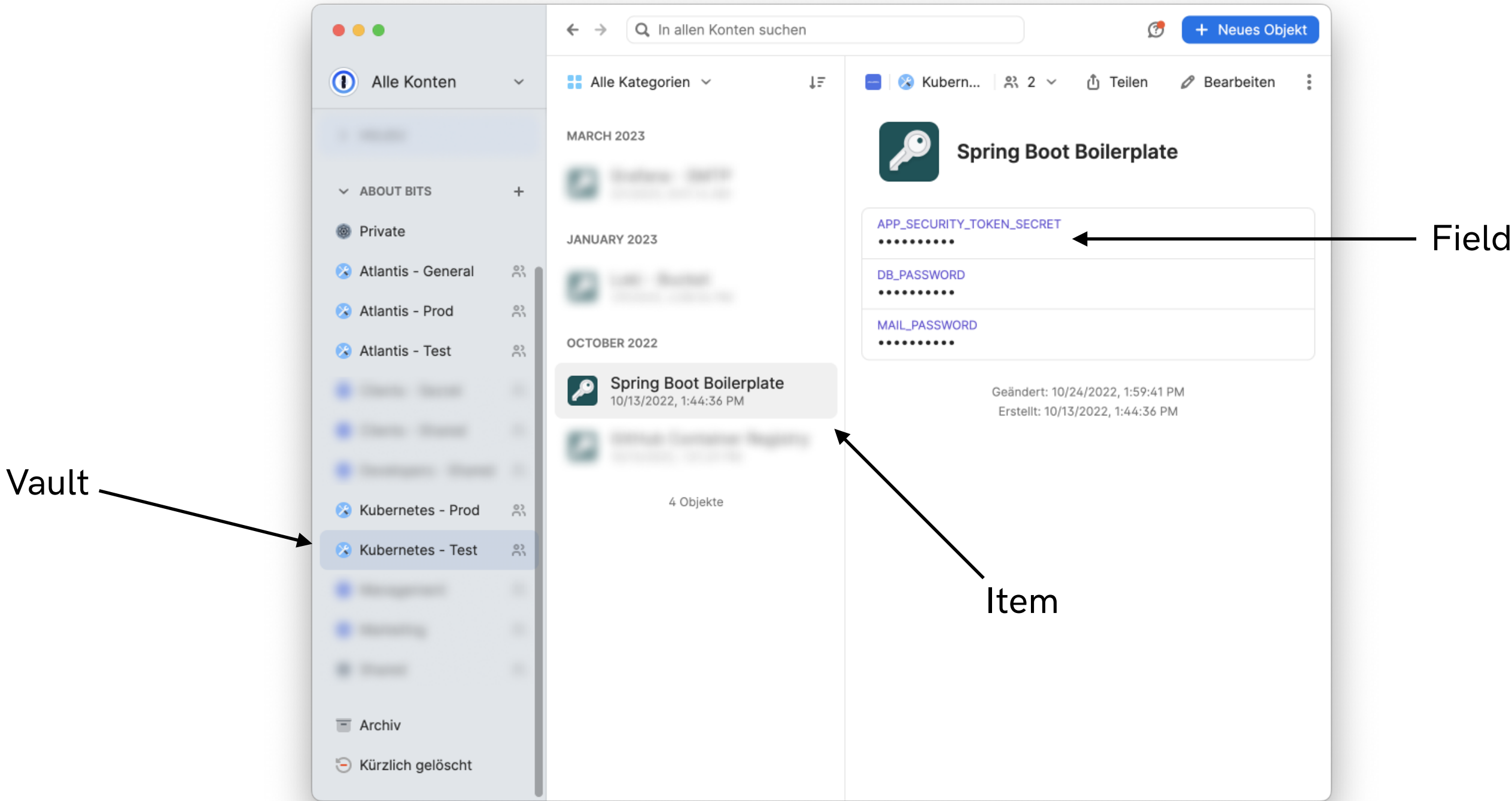
1Password Connect Kubernetes Operator

Provides the ability to integrate Kubernetes with 1Password



Kubernetes - Secrets Management

Step 1: Create an item in 1Password with the required secrets



Kubernetes - Secrets Management

Step 2: Create a OnePasswordItem pointing to the 1Password item

```
apiVersion: onepassword.com/v1
kind: OnePasswordItem
metadata:
  namespace: boilerplate
  name: app-secrets
spec:
  itemPath: "vaults/Kubernetes - Test/items/Spring Boot Boilerplate"
```

Kubernetes - Secrets Management

Step 3: The 1Password Operator automatically creates the according secret

```
kind: Secret
apiVersion: v1
metadata:
  name: app-secrets
  namespace: boilerplate
data:
  APP_SECURITY_TOKEN_SECRET: xxx
  DB_PASSWORD: xxx
  MAIL_PASSWORD: xxx
type: Opaque
```

Kubernetes - Secrets Management

Step 4: Use the secret in a deployment

```
apiVersion: apps/v1
kind: Deployment
metadata:
  namespace: boilerplate
  name: app
spec:
  ...
  template:
    ...
    spec:
      containers:
        - name: app
          ...
          env:
            - name: DB_USERNAME
              value: spring-boot-boilerplate
            - name: DB_PASSWORD
              valueFrom:
                secretKeyRef:
                  name: app-secrets
                  key: DB_PASSWORD
            - name: MAIL_USERNAME
              value: info@aboutbits.it
            - name: MAIL_PASSWORD
              valueFrom:
                secretKeyRef:
                  name: app-secrets
                  key: MAIL_PASSWORD
```

1 Password Family



First year for free! 🎉

Questions?

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